

PerturbVAE: A Leakage-Aware Benchmark for Single-Cell Perturbation-Response Prediction

Brhanu F. Znabu
University of Nebraska–Lincoln

June 2026

Abstract

PerturbVAE is a conditional variational autoencoder that predicts the transcriptional response to a genetic perturbation it has never seen, evaluated under three leakage-controlled held-out splits on two real Perturb-seq datasets (Norman et al. 2019; Replogle et al. 2022). The focus of the project is the evaluation rather than the architecture, which is deliberately small and standard: how much of the apparent ability to predict perturbation responses survives when the held-out perturbations are not allowed to be functional neighbors of the training perturbations, and where a learned model adds value over a baseline that ignores perturbation identity. The result is mixed and reported as such. The learned model beats a strong mean-of-training baseline in one regime, compositional prediction of unseen *double* perturbations (delta-Pearson 0.748 vs 0.551 on Norman), and loses to it on every single-perturbation split of both datasets. A leakage-free, function-grouped split changes the measured generalization substantially on a diverse dataset (-0.27 Pearson on Norman) and barely at all on a homogeneous one (-0.02 on Replogle), so the leakage gap is real but dataset-dependent. The contribution is the benchmark design and the honest cross-dataset accounting.

1 Summary of results

Question	Answer
Does the cVAE beat a strong baseline anywhere?	Yes, on compositional generalization. Predicting unseen <i>double</i> perturbations from singles (Norman), it reaches delta-Pearson 0.748 vs the mean-of-training baseline’s 0.551 (+0.20), and nearly doubles the top-20 DE-gene overlap (0.34 vs 0.20).
Does it beat the baseline on single-perturbation prediction?	No. On every single-perturbation split of both datasets the mean-of-training baseline is higher. The advantage is specific to composition, not single-gene response.
Is there a leakage gap between a random and a leakage-free split?	Yes, but dataset-dependent. Large on Norman (cVAE -0.27 , baseline -0.20); nearly absent on Replogle (≈ -0.02). A random-split number can overstate generalization by up to 0.2 to 0.27 Pearson.
Is the evaluation honest about scale?	Yes. Two datasets, three split designs, deterministic hashing, a zero-effect control and a strong mean baseline on every split, and DE-gene overlap reported alongside correlation.

2 Motivation

Perturb-seq couples a CRISPR perturbation to single-cell RNA-seq, so each cell carries both which gene was hit and what the transcriptome did about it. Models that predict the response to an unseen perturbation are attractive because the combinatorial space of perturbations, and especially of *combinations* of perturbations, is far too large to measure exhaustively. The field reports such models with delta-correlation against held-out perturbations. Two things make those numbers easy to over-read, and both are the subject of this project.

The baseline is unusually strong. Most perturbations share a large common component, a stress / growth-arrest / cell-cycle signature, so predicting the average training perturbation, ignoring which gene was perturbed, already correlates respectably with held-out responses. A learned model has to clear that bar to be useful [5, 6].

The held-out perturbations leak. A random train/test split routinely puts a gene’s pathway partners or complex members in the training set, so the model can interpolate to the held-out gene from a functional neighbor it has already seen. This is the same failure mode as phylogenetic leakage in sequence models: the test point is not actually novel. The relevant question is how the model does when *whole functional groups* are withheld. PerturbVAE makes both unavoidable: a zero-effect control and a mean-of-training baseline on every split, and a leakage-free, function-grouped split next to the usual random one.

3 Data

Both datasets are loaded through the GEARS `PertData` processed `AnnData` and used exactly as distributed (log-normalized expression, conditions in the `GENE+ctrl` / `GENEA+GENEB` / `ctrl` convention). No synthetic or simulated data is used anywhere in the results; the only synthetic arrays in the repository are inside `tests/`, explicitly as correctness checks.

Dataset	Cells	Genes	Single conds	Double conds
Norman 2019 (K562, combinatorial)	91,205	5,045	152	131
Replogle 2022 (K562, essential-gene)	162,751	5,000	1,092	0

Norman is the compositional dataset: it contains genuine two-gene perturbations, which is what makes split C possible. Replogle is a much larger single-perturbation screen with no combinations, used to test whether the single-perturbation findings replicate at roughly seven times the number of perturbations and on a different experimental design. The prediction target is the **pseudobulk delta**: for each condition, the mean expression over its cells minus the mean over non-targeting control cells.

4 Evaluation design and contribution

Three held-out splits, each isolating a different kind of generalization, all deterministic (BLAKE2b hashing of condition / group keys) so the splits are identical across machines and runs.

- **A, random singles.** A random 80/20 split over single-gene conditions; the optimistic number matching common practice.
- **B, function-grouped singles (leakage-free).** Genes are clustered into functional groups from their control-cell co-expression signature, and *entire groups* are held out, so no gene from

a held-out group ever appears in training. Groups are selected smallest-first by deterministic hash until $\sim 20\%$ of perturbed genes are withheld. The gap between A and B is the **leakage gap**.

- **C, combinatorial.** Train on all single perturbations, test on all *double* perturbations (Norman only). A fixed mean vector structurally cannot predict an interaction, so this is the split where a learned conditional model can outperform the baseline.

Each held-out perturbation is scored by delta-Pearson, delta-MSE, and the Jaccard overlap of the top-20 differentially expressed genes; we report the mean across held-out conditions. To predict a perturbation never seen in training, each gene is represented by a **generalizable embedding**, the PCA of its gene-gene co-expression signature on control cells, defined for every gene. A condition’s embedding is the mean of its perturbed genes’ embeddings, which is also what lets a double perturbation be embedded from two singles in split C.

5 Model

A conditional VAE on the per-cell delta (cell expression minus the control mean): an encoder $q(z | x)$ from the observed delta, a prior $p(z | e)$ from the perturbation embedding e , and a decoder $\hat{x}(z, e)$ conditioned on *both* the latent and the perturbation embedding. Conditioning the decoder on the perturbation is what lets one trained model produce a different response for a different (including unseen) perturbation. Because the model is trained on deltas, the predicted response is read off directly by sampling z from the perturbation prior and decoding, averaged over draws. It is deliberately small: 64-d latent, 512-d hidden, MSE reconstruction plus KL to the perturbation prior, β -annealed over the first third of training, 40 epochs, Adam 10^{-3} , on a single L40S / H100. The model is held fixed across all splits.

6 Baselines

control-mean predicts zero change for every perturbation (its delta-Pearson is undefined, so it appears as an MSE floor). **mean-of-training** predicts the average training-set delta for every test perturbation, ignoring which gene was perturbed; this is the strong baseline that captures the shared stress/growth component dominating most responses.

7 Results

7.1 Per-split performance

Dataset	Split	n test	cVAE r	MoT r	cVAE Jacc.	MoT Jacc.
Norman	A random	20	0.522	0.616	0.243	0.266
Norman	B function	28	0.257	0.412	0.125	0.173
Norman	C combinatorial	131	0.748	0.551	0.341	0.199
Replogle	A random	237	0.305	0.391	0.084	0.072
Replogle	B function	86	0.283	0.370	0.082	0.076

Finding 1: the compositional win. On Norman’s combinatorial split the cVAE reaches delta-Pearson 0.748 vs 0.551 for mean-of-training (+0.197), and the top-20 DE-gene overlap rises

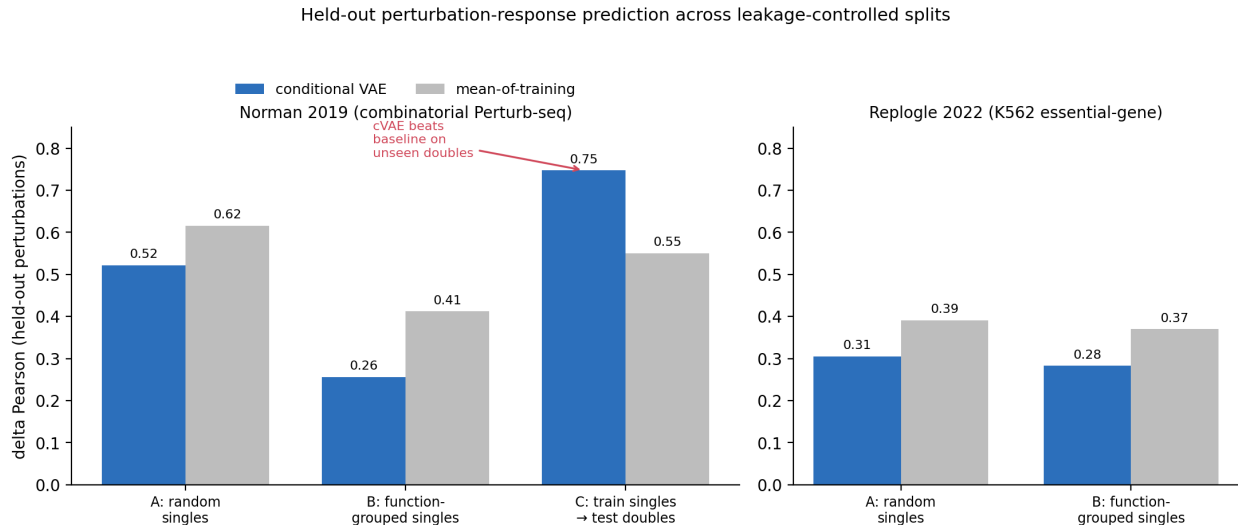


Figure 1: Mean delta-Pearson over held-out perturbations, conditional VAE (blue) vs mean-of-training (grey). Left: Norman, three splits. Right: Replogle, two splits. The learned model exceeds the strong baseline only on the combinatorial split.

from 0.199 to 0.341. This is the one regime where the prediction is genuinely about *interaction*: the joint effect of two perturbed genes is not the average training delta, and a model that composes two single-gene embeddings captures structure a fixed vector cannot. This is the project’s positive result.

Finding 3: single-perturbation calibration (honest negative). On every single-perturbation split of both datasets (A and B, Norman and Replogle), the mean-of-training baseline is higher than the cVAE. A small VAE trained on a few hundred to a thousand perturbations does not beat predicting the average response when the test perturbation is itself a single gene, consistent with the well-documented difficulty of beating simple baselines on single-perturbation benchmarks [5, 6]. It is reported here rather than hidden by quoting only the combinatorial number.

7.2 The leakage gap

Dataset	Method	A random	B function	gap (A–B)
Norman	cVAE	0.522	0.257	0.265
Norman	mean-of-training	0.616	0.412	0.204
Replogle	cVAE	0.305	0.283	0.023
Replogle	mean-of-training	0.391	0.370	0.021

Finding 2: the leakage gap is real but dataset-dependent. On Norman, withholding whole functional groups instead of random genes drops delta-Pearson by 0.265 for the cVAE and 0.204 even for the mean-of-training baseline: a fifth to a quarter of the apparent random-split performance was interpolation between functionally similar perturbations, and some of it disappears even for a model that never looks at gene identity, because random splits also leak through which conditions land in the test mean. On Replogle the same procedure moves the number by only ~ 0.02 . The reason is informative rather than contradictory: Replogle is a 1,092-gene essential-gene screen whose responses are dominated by a single shared growth-arrest signature, so per-perturbation

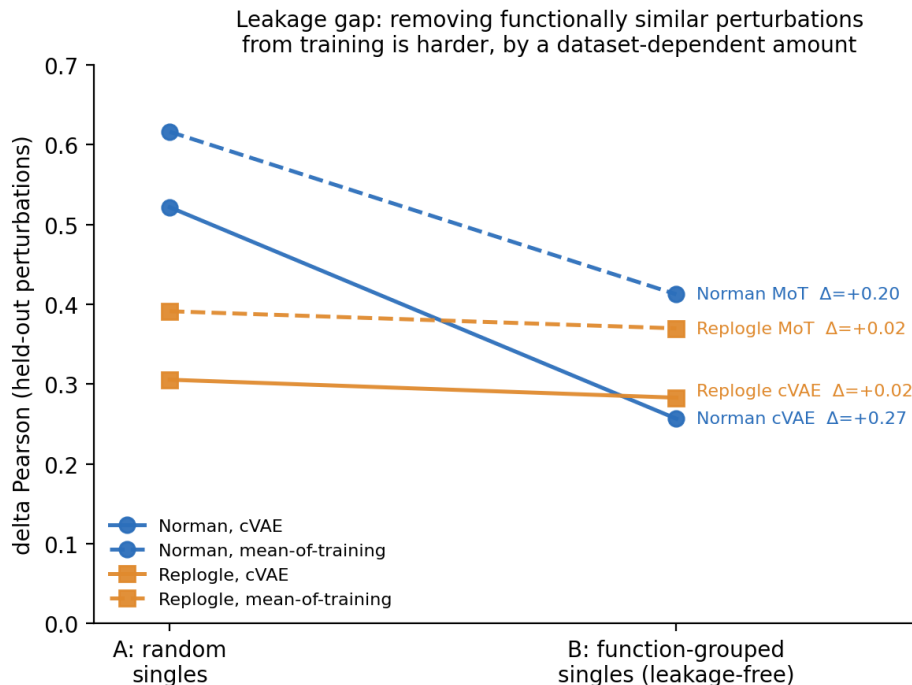


Figure 2: Delta-Pearson on the random split (A) vs the leakage-free function-grouped split (B). The slope is the leakage gap: large on Norman and almost flat on Replogle, for both the learned model and the baseline.

deltas are relatively homogeneous, the mean baseline is already near the ceiling, and removing functional neighbors barely changes what is predictable. The leakage gap is therefore not a universal constant; its size tracks how much *perturbation-specific* structure a dataset contains. The practical consequence stands: on a diverse perturbation set a random-split number can overstate leakage-free generalization by 0.2 to 0.27 Pearson, and a leakage-free split is necessary to know which.

8 What this shows, and what it does not

It shows: (i) a learned conditional model can beat a strong, identity-blind baseline specifically on compositional prediction of unseen perturbation combinations; (ii) on single-perturbation prediction that same model does not beat the baseline at this scale; (iii) a leakage-free, function-grouped split changes measured generalization substantially on a diverse dataset and barely at all on a homogeneous one. Together these calibrate *when* perturbation-response prediction is hard and *where* a learned model helps.

It does not show that this particular cVAE is a competitive perturbation model. The architecture is intentionally small and untuned; larger conditional models (GEARS, scGPT-style, CPA) would likely raise all of these numbers. The functional groups are derived from co-expression rather than curated pathway/complex annotations, a reproducible but coarse proxy for a functional neighbor. And the evaluation is pseudobulk delta-correlation, which rewards getting the dominant direction right and is forgiving about the tails. The contribution is the benchmark design and the honest cross-dataset accounting, not a state-of-the-art model.

9 Reproducibility

Every number is produced by the scripts in the repository from the public datasets with deterministic splits:

```
python src/scripts/run_baselines.py --h5ad data/norman/perturb_processed.h5ad \  
    --out results/baselines_norman.json  
python src/scripts/train.py --h5ad data/norman/perturb_processed.h5ad \  
    --out results/cvae_norman_delta.json --epochs 40  
# repeat both with data/replogle_k562_essential/perturb_processed.h5ad  
python src/scripts/make_figures.py
```

`tests/test_smoke.py` checks the splits, metrics, baselines, and model on small synthetic arrays. These are correctness tests, explicitly labeled as such, not part of any reported result.

References

- [1] Norman, T.M., Horlbeck, M.A., Replogle, J.M., et al. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. *Science* 365(6455):786–793, 2019.
- [2] Replogle, J.M., Saunders, R.A., Pogson, A.N., et al. Mapping information-rich genotype–phenotype landscapes with genome-scale Perturb-seq. *Cell* 185(14):2559–2575, 2022.
- [3] Roohani, Y., Huang, K., Leskovec, J. Predicting transcriptional outcomes of novel multigene perturbations with GEARS. *Nature Biotechnology* 42:927–935, 2024.
- [4] Lotfollahi, M., Klimovskaia Susmelj, A., De Donno, C., et al. Predicting cellular responses to complex perturbations in high-throughput screens (CPA). *Molecular Systems Biology* 19(6):e11517, 2023.
- [5] Ahlmann-Eltze, C., Huber, W., Anders, S. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods* 22:1657–1661, 2025.
- [6] Kernfeld, E., Yang, Y., Weinstock, J.S., Battle, A., Cahan, P. A systematic comparison of computational methods for expression forecasting. *bioRxiv* 2023.07.28.551039, 2023.